

Readme Contents

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1. What is this?

The Bcalc Firearm Management App is an open-source firearm, optics, and ammunition tracker with no cloud-based services or subscriptions.

You can enter multiple configurations of firearms, optics, and ammunition. For each configuration, you can track range session performance and conditions. Data reports such as rounds per firearm and top configurations by caliber are available.

An internet connection is not required once installed. No user data is transmitted.

2. Who is this for?

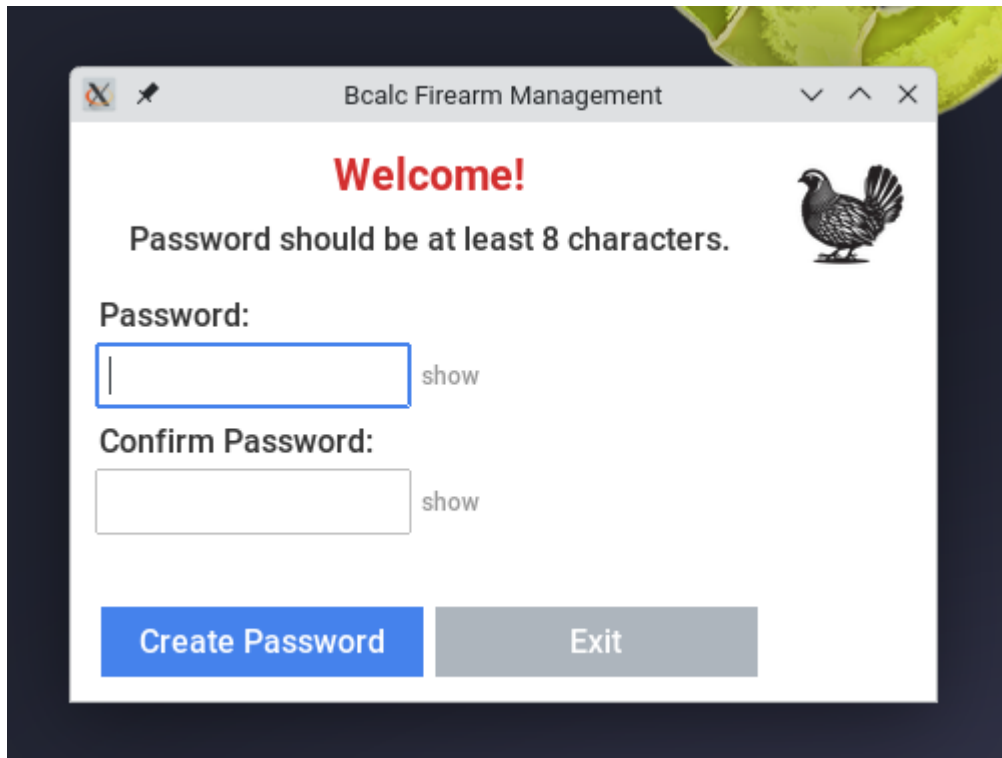
Hunters and hobbyists who prefer to use open-source applications. It's easy to track your firearm performance with different ammunition, whether purchased or hand-load.

3. Basic Linux Installation Guide

1. Linux (standalone application – no python installation required)
 - download "bcalctracker" from the <https://github.com/briancalc/bcalctracker/releases> page
 - to run using file manager, double-click bcalctracker
 - to run from the terminal `./bcalctracker`
2. Windows
 - in progress
3. Mac
 - in progress

4. User Guide

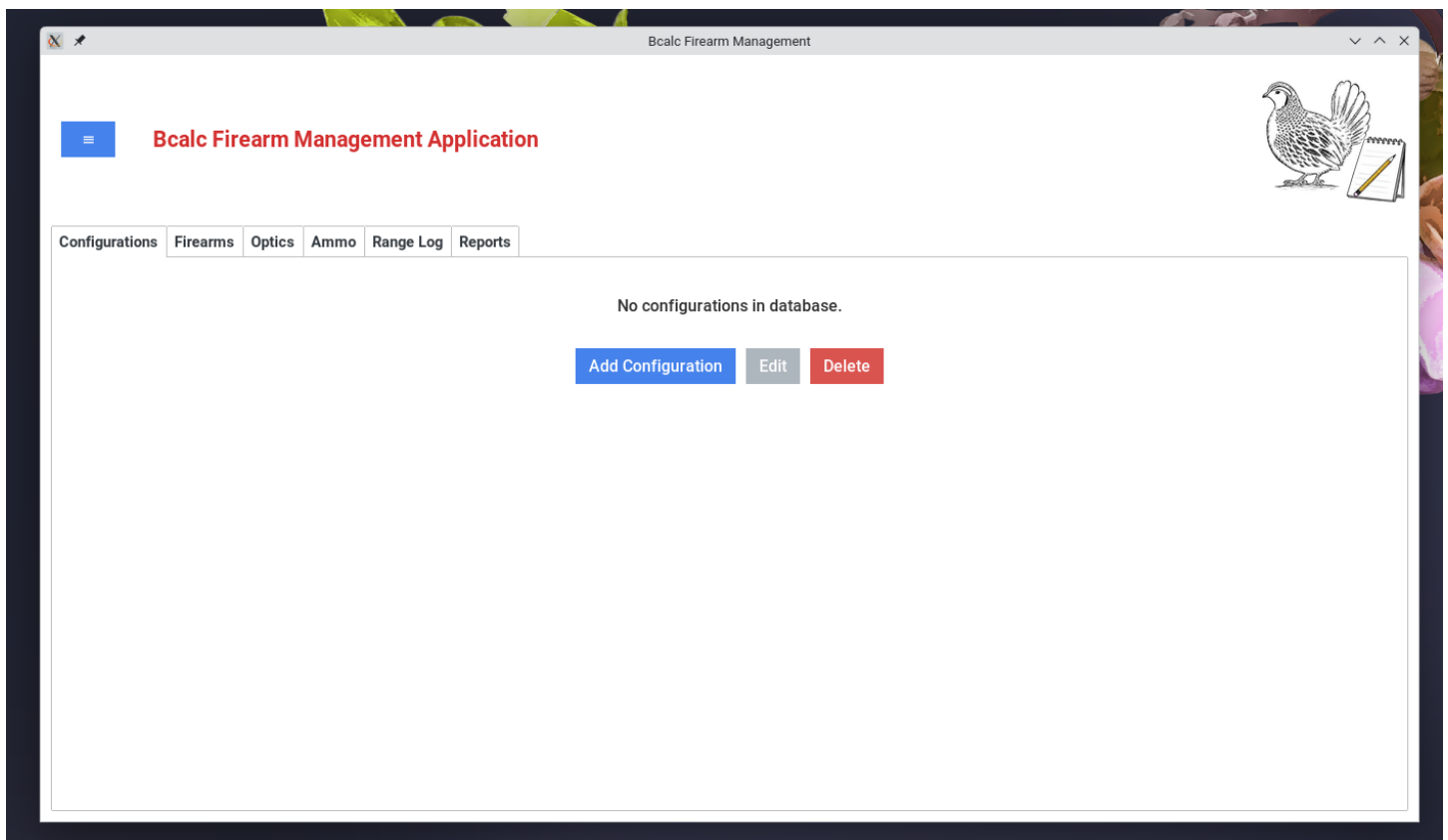
Step 1. Create a password to secure your records
Don't misplace your password as there is no recovery option.



The screenshot shows a web browser window titled "Bcalc Firearm Management". The page has a white background with a dark blue header bar. In the top right corner of the header, there is a small icon of a quail. The main content area features a large red "Welcome!" message. Below it, a black text prompt states "Password should be at least 8 characters." To the right of this text is a small quail icon. The form consists of two input fields: "Password:" and "Confirm Password:". Each field has a "show" link to its right. At the bottom, there are two buttons: a blue "Create Password" button and a grey "Exit" button.

Step 2: Firearms

Configurations are what facilitate the collection of range session data. Configurations can consist of a firearm, an optic, and a specific ammunition or, at the minimum, just a firearm.



The screenshot shows the main dashboard of the "Bcalc Firearm Management Application". The header bar is dark blue with a white hamburger menu icon on the left and a quail icon on the right. The main content area has a white background. At the top left, there is a blue button with a white hamburger menu icon. Below it, the text "Bcalc Firearm Management Application" is displayed in red. A horizontal navigation bar contains several tabs: "Configurations", "Firearms", "Optics", "Ammo", "Range Log", and "Reports". The "Configurations" tab is currently selected. The main content area displays the message "No configurations in database." in the center. Below this message are three buttons: a blue "Add Configuration" button, a grey "Edit" button, and a red "Delete" button.

So, you need to enter a firearm: click on the Firearms tab and select the blue Add Firearm button. Firearm Name, Caliber Primary, and Barrel Length are required fields. Enter as many of the other fields as you would like, including attaching a photo. You can always return and edit the information later.

The screenshot displays the 'Bcalc Firearm Management' application window. A modal form titled 'Add New Firearm' is open, allowing users to input details for a new firearm. The form is organized into several sections: 'Basic Information' with fields for Firearm Name, Manufacturer, Type, Serial #, Caliber Primary, Caliber Secondary, and Barrel Length; 'Purchase Details (optional)' with fields for Purchase Date, Purchase Location, and Purchase Cost; 'Sold Status' with a checkbox for 'Mark as Sold'; and 'Photo & Notes' with a Firearm Photo upload field and a Notes field. The 'Firearm Name' field is populated with 'BAR Mk3 Stalker', 'Manufacturer' with 'Browning', 'Type' with 'Rifle: Semi', 'Serial #' with 'BAR Mk3 Stalker', and 'Caliber Primary' with '243 Win'. The 'Barrel Length (in):' field is empty. The 'Purchase Date' field is populated with 'YYYY-MM-DD', 'Purchase Location' is empty, and 'Purchase Cost' is empty. The 'Mark as Sold' checkbox is unchecked. The 'Firearm Photo' field is empty, and the 'Notes' field is empty. The 'Browse...' button is visible next to the 'Firearm Photo' field. At the bottom of the modal, there are 'Save' and 'Cancel' buttons. Below the modal, in the background application window, there are 'Add Firearm', 'Edit', and 'Delete' buttons. The application window also shows a sidebar with 'Configurations' and a list of configurations including 'Name', 'Henry H9 Provi', and 'American Gen1'. The application title bar is 'Bcalc Firearm Management'.

Add New Firearm

Basic Information

Firearm Name: BAR Mk3 Stalker Manufacturer: Browning

Type: Rifle: Semi Serial #: BAR Mk3 Stalker

Caliber Primary: 243 Win Caliber Secondary:

Barrel Length (in):

Purchase Details (optional)

Purchase Date: YYYY-MM-DD Purchase Location:

Purchase Cost:

Sold Status

☐ Mark as Sold

Photo & Notes

Firearm Photo: Browse...

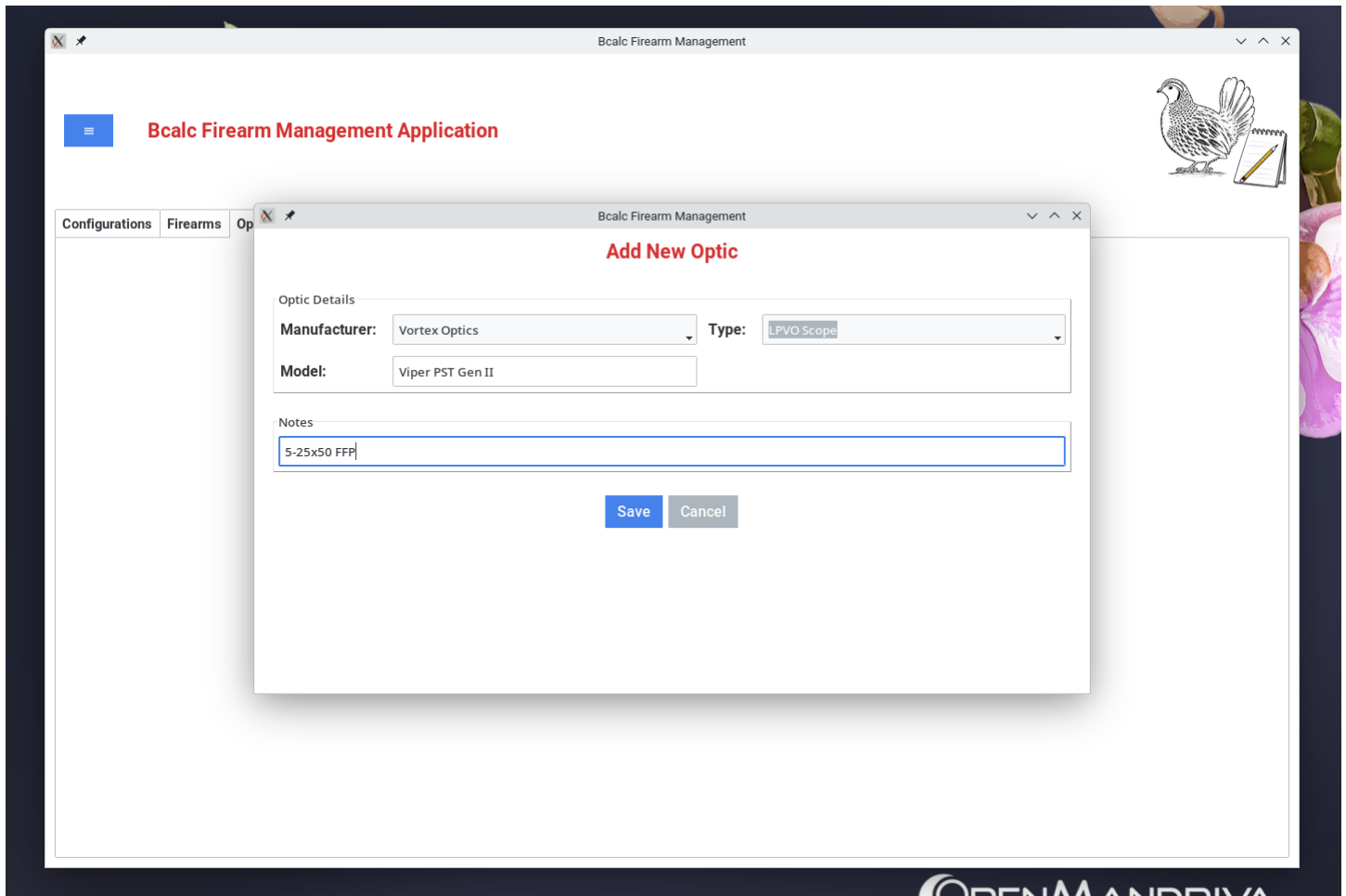
Notes:

Save Cancel

Add Firearm Edit Delete

Step 3. Optics

Next, you can enter an optic. Select the Optic Tab and then the Add Optics button.



The screenshot displays the Bcalc Firearm Management Application interface. The main window has a title bar "Bcalc Firearm Management" and a header area with a menu icon and the text "Bcalc Firearm Management Application". A sidebar on the left contains tabs for "Configurations", "Firearms", and "Optics". The "Optics" tab is selected. A modal dialog box titled "Add New Optic" is open in the center. It contains the following fields:

- Optic Details**
 - Manufacturer:** A dropdown menu with "Vortex Optics" selected.
 - Type:** A dropdown menu with "LPVO Scope" selected.
 - Model:** A text input field containing "Viper PST Gen II".
- Notes**
 - A text input field containing "5-25x50 FFP".

At the bottom of the dialog box are two buttons: "Save" (blue) and "Cancel" (grey). The background of the application window shows a faint illustration of a turkey and some flowers.

Step 4. Ammunition

You guessed it. Enter ammunition by selecting the Ammo Tab and then the Add Ammo button. Both purchased and hand-load ammunition can be tracked.

Bcalc Firearm Management

Add New Ammo Record

Basic Information

Manufacturer: Caliber/ Gauge:

Grains (Proj.): Projectile Type:

Shot Size: # of Pellets:

Ammo Box Name: Use Case:

Drag Function: Ballistic Coeff:

☐ Do Not Rebuy (Poor Performance)

Reloading Components (Optional)

Bullet Mfg: Powder Mfg:

Bullet Name: Powder Name:

Brass Mfg: Powder Lot:

Case Weight: Powder Grains:

Overall Length: Primer Mfg:

CBTO: Primer Type:

Crimp Type: Primer Lot:

Hull Mfg: Wad Mfg:

Batch Stamps

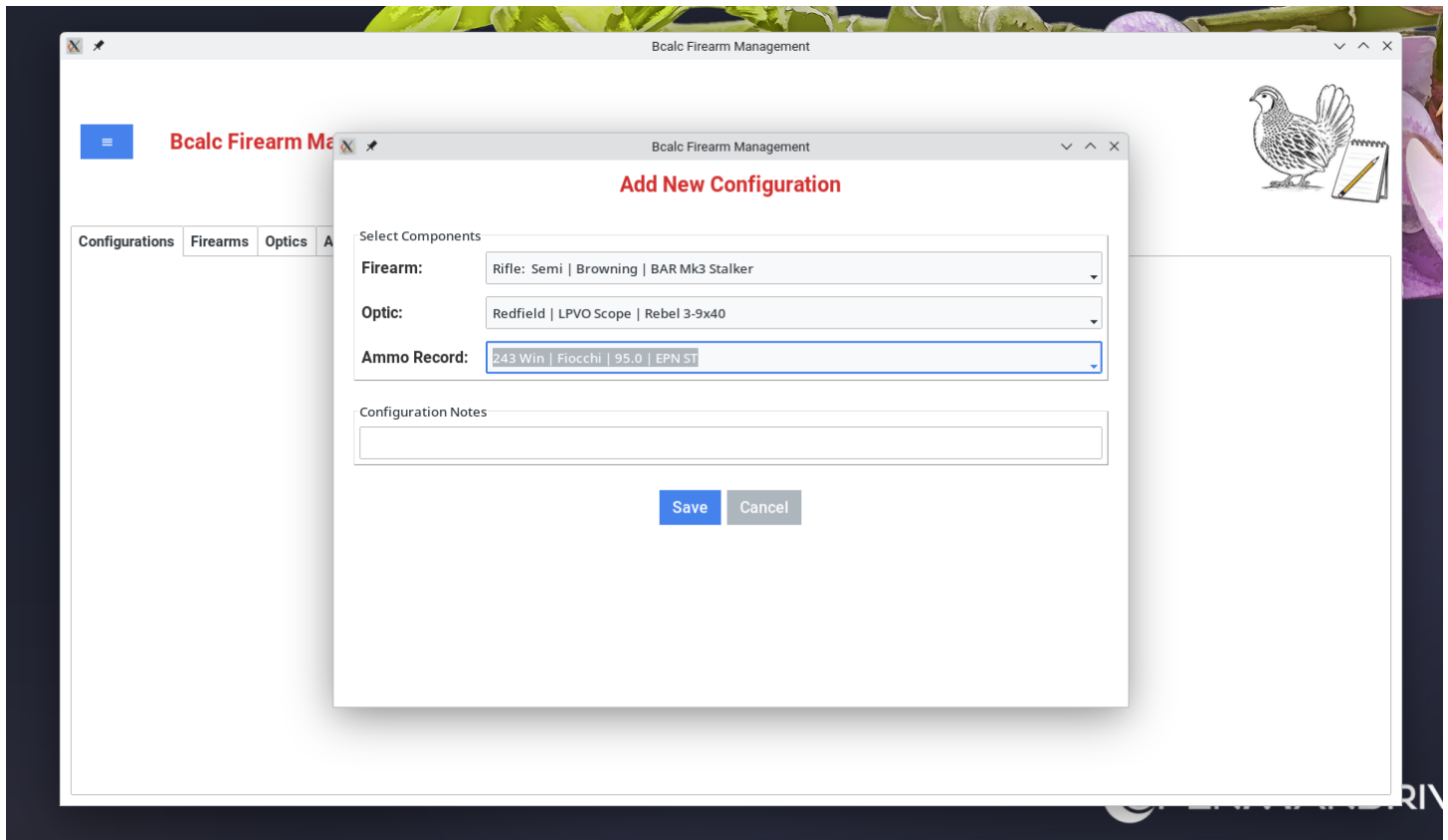
Batch Quantity: Date Loaded (YYYY-MM-DD):

Notes

Save **Cancel**

Step 5. Configurations

Enter a configuration by selecting the Configuration Tab and then the Add Configuration button. At a minimum, a configuration must include a firearm. There are no practical limits to the number of different configurations of firearms, optics, and ammunition. It is very likely that each firearm will have several ammo brands associated with it.



The screenshot displays the Bcalc Firearm Management application interface. A modal dialog box titled "Add New Configuration" is open, allowing users to create a new setup. The dialog includes the following fields:

- Select Components:** A section containing three dropdown menus:
 - Firearm:** Currently set to "Rifle: Semi | Browning | BAR Mk3 Stalker".
 - Optic:** Currently set to "Redfield | LPVO Scope | Rebel 3-9x40".
 - Ammo Record:** Currently set to "243 Win | Fiocchi | 95.0 | EPN ST".
- Configuration Notes:** A text area for additional information.
- Buttons:** "Save" and "Cancel" buttons at the bottom.

The background application window shows a sidebar with tabs for "Configurations", "Firearms", "Optics", and "Ammunition". The "Configurations" tab is active. The application title bar reads "Bcalc Firearm Management". A small illustration of a quail is visible in the top right corner of the application window.

Step 6. Range Log

Select the Range Log Tab and the add Session button. Configurations are the primary key for the range results. All other data is optional. A Rating field is provided to help track the firearm-ammo performance.

Bcalc Firearm Management

Add New Range Session

Session Details

Configuration:

Date (YYYY-MM-DD): Location:

Performance & Condition Data

Rounds Fired:	80	Target Range:	100	Wind Speed:	
Avg Muzzle Vel:	3240	Avg MOA:	1.4	Wind Dir (0-359):	
Max Muzzle Vel:	3260	Best MOA:	1.0	Temperature:	
Std Dev Vel:		Std Dev MOA:		Humidity:	
Ext Spread Vel:		Ext Spread MOA:		Rating:	unknown

Photo & Notes

Target Photo:

Notes:

Step 7. Reports
Select the Reports Tab. Several data points are shown including total rounds per firearm and the top configurations per caliber calculated using MOA and Std Dev MOA data.

Bcalc Firearm Management Application

ConfigurationsFirearmsOpticsAmmoRange LogReports

Firearm Round Stats

Select Firearm:

Browning BAR Mk3 Stalker

Total Rounds Fired: 195

Top Ammo: Fiocchi EPN ST (243 Win)

Top Configurations by Caliber

Select Caliber:

243 Win

Rank	Configuration	Rating	Rounds
1	Browning BAR Mk3 Stalker Redfield Reb	No Rating	195
2	Ruger American Gen II Predator Vortex	No Rating	85

Configuration Stats

Select Configuration:

Henry H9 Provider | Leupold | Hornady

Avg Velocity: 2465 fps

Avg MOA: 1.50

Ammo Manufacturer & Caliber Analysis

Manufacturer:

Fiocchi

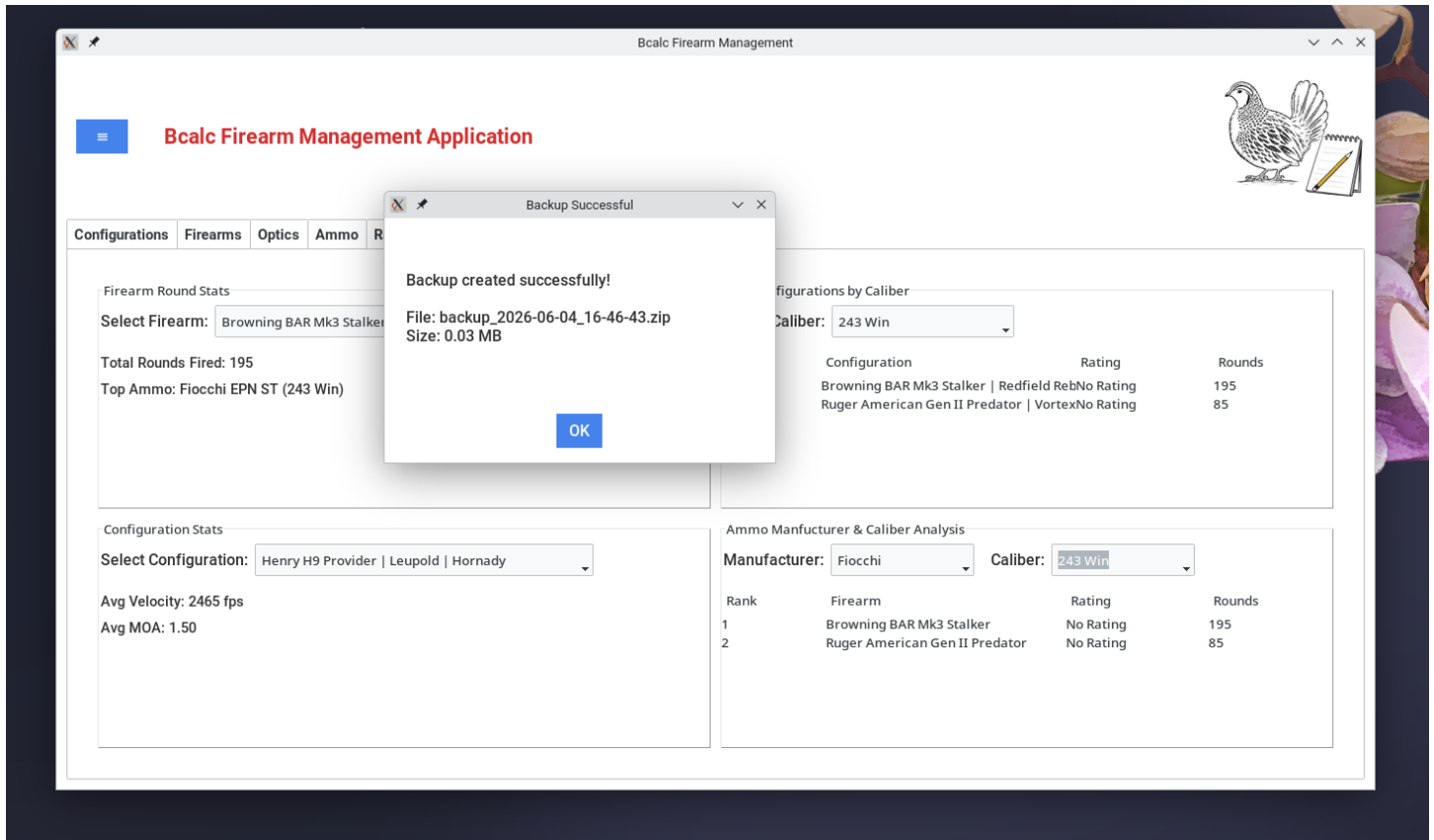
Caliber:

243 Win

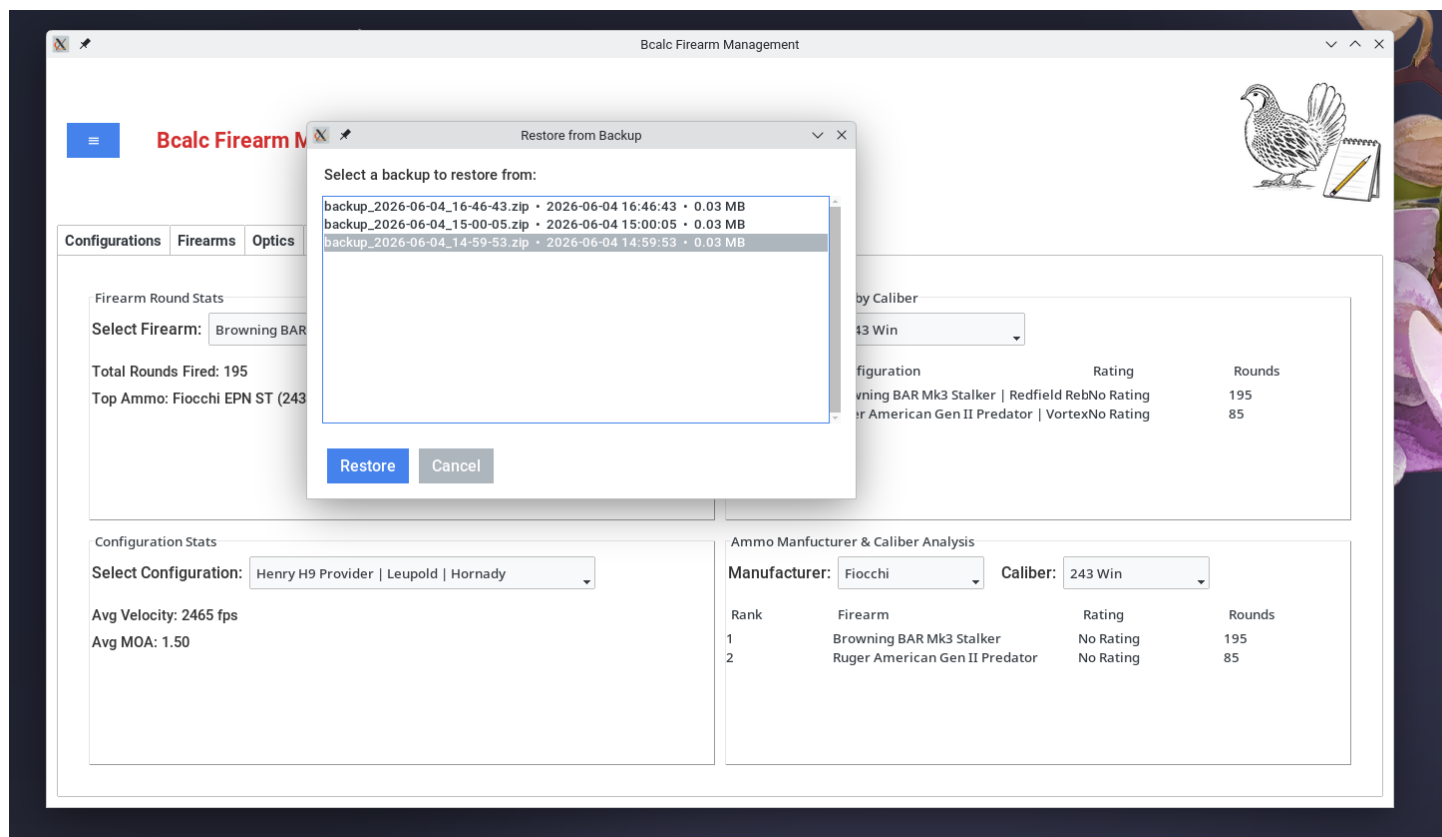
Rank	Firearm	Rating	Rounds
1	Browning BAR Mk3 Stalker	No Rating	195
2	Ruger American Gen II Predator	No Rating	85

Step 8. Optional: Backup and Restore

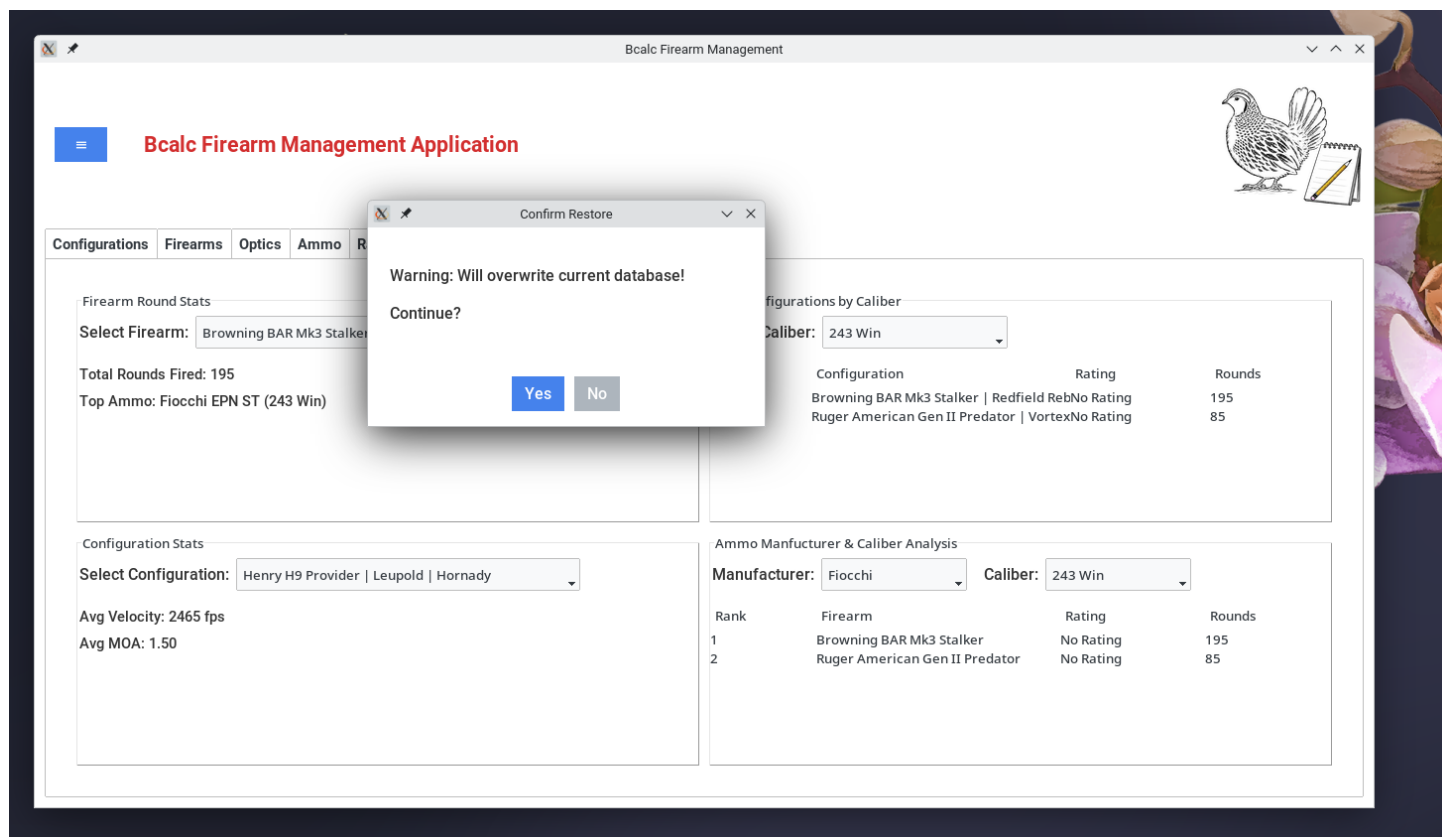
It's a good idea to periodically save a copy of the database just in case. Click the blue hamburger menu at the top left of the window and select Backup Database.



To restore your records from a backup, click the blue hamburger menu and select Restore Database. Pick a backup and press Restore.



Your current database will be replaced by the backup so be certain you want to restore your previous records.



5. Credits & License

The Bcalc Firearm Management App was created by Brian Calc.

GNU General Public License (GPL) Version 3.